

ELARA: Energy-Efficient and Latency-Aware Service Routing with Adaptive Replica Redeployment for Microservices in Space Computing Power Networks

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Abstract—The rapid advancement of the Space Computing Power Network (SCPN), comprising numerous low Earth orbit (LEO) satellites, has enabled the execution of complex in-orbit observation and computation tasks. To facilitate this, microservice architecture decomposes large, monolithic remote sensing and image processing applications into independent, schedulable, and reusable services deployed in a distributed manner across LEO satellites. However, integrating these capabilities presents significant challenges arising from the limited computing resources, constrained power provisions of individual satellites, and the dynamic nature of space network topology and channel conditions. This paper addresses these challenges by proposing an energy-aware service routing algorithm for microservices within the SCPN. The algorithm aims to minimize the energy consumed by the procedures of processing and transmitting service data in both computation and communication. By modeling the service routing problem as a Directed Steiner Tree (DST) and implementing heuristic-based solutions, our algorithm effectively responds to the diverse service patterns requested by terrestrial clients. Comprehensive experimental results demonstrate that our approach significantly outperforms existing methods in terms of energy efficiency, robustness, stability, and adaptability.

Index Terms—Energy, Microservice, Routing, Space Network

I. INTRODUCTION

In recent years, large-scale LEO satellite networks have garnered widespread attention as one of the recognized highlights of 6G technology [1]. Modern LEO mega-constellations, consisting of hundreds to thousands of LEO satellites connected by inter-satellite links (ISLs), enable the globally ubiquitous connections [2]. Meanwhile, advancements in remote sensing and chip technology have enabled modern LEO satellites to be equipped with high-performance imaging and intelligent computing payloads [3], [4], thus supporting ubiquitous intelligent services such as weather forecasting, environmental monitoring, and national security [5], [6]. Hundreds to thousands of satellite computing nodes collectively form the SCPN, where the strategic allocation of sensing, communication, and computing resources to efficiently fulfill diverse intelligent service requests has emerged as a critical challenge in the context of 6G integrated space-air-ground networks [7].

Particularly, due to the large-scale deployment of LEO satellites, advancements in satellite remote sensing and data analytics technology, and the high image resolution afforded by the lower orbital altitude, LEO satellite remote sensing

services have demonstrated unprecedented momentum. These services continuously generate massive amounts of data featuring multiple formats and modalities, catering to extensive user demands [8]–[10]. This paper illustrates these application scenarios as depicted in Fig. 1.

Due to the enormous and continuously growing volume of remote sensing data, downloading all satellite data to the ground for centralized processing consumes substantial satellite-to-ground communication resources and may cause congestion for other services. Therefore, leveraging onboard computational resources to perform partial or complete processing of remote sensing tasks and then downloading the computational results to the ground can save communication resources and mitigate privacy and security risks associated with downloading raw data [11]. Furthermore, considering the potential computational resource constraints of a single satellite, the multi-satellite collaborative computing paradigm has emerged as a mainstream research focus [12], [13]. Specifically, thanks to the development of containerization technologies and the evolution of microservice architectures [14]–[16], the functional modules of the user applications, which are traditionally deployed in terrestrial data centers to process raw data postback from satellites, [17], can now be decomposed into stateless microservices and packaged as independent and isolated containers [18]. These services can then be dispatched to computing nodes of the SCPN for on-board computation so as to reduce the transmission delay caused by limited bandwidth and vulnerable channel conditions between ground and space networks [19], [20].

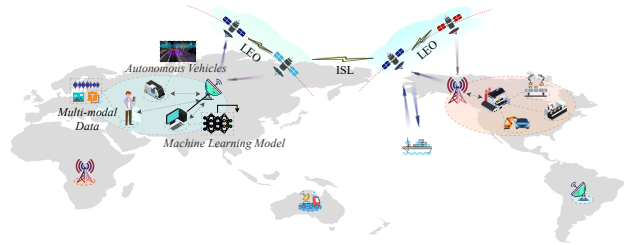


Fig. 1: Space and terrestrial network topology.

However, unlike terrestrial computing nodes with adequate computing resources and power supply. The SCPN nodes present unique limitations, i.e., constrained computing re-

sources and poor power supply. On the one hand, satellites can only be equipped with limited computing and storage resources as a consequence of the constraints of size and weight, as well as the damage risk caused to circuit boards by intense radiation in space [21]. On the other hand, the energy supply for satellites is primarily derived from solar energy harvesting, which is utilized for various satellite functions, including attitude control, transceivers, cameras, and processors. Furthermore, considering the area of solar panels, the efficiency of the materials used for energy harvesting, and hardware degradation, the battery power available for satellites is extremely limited compared to the energy supply available to terrestrial servers. [22], [23].

Hence, different from terrestrial settings [24]–[26], the energy efficiency should be given high priority and regarded as the main concern when devising the routing schemes in the SCPN [22], [27], [28]. To this end, this work presents an energy-aware service routing algorithm for microservices in the SCPN, aiming to minimize energy consumption in both computation and communication to meet the task requirements of terrestrial clients. Specifically, we first model the service routing problem as a Directed Steiner Tree (DST) on the directed graph, which is a classic combinatorial optimization problem and has been proved to be NP-hard [29], [30], then design a series of heuristic-based algorithms to obtain the approximate optimal solution for the service routing problem in the SCPN.

Our main contributions are summarized as follows:

- **Service Routing Modeling in the SCPN:** We identify satellite energy consumption as a critical bottleneck of microservice routing in the SCPN. To this end, we meticulously model the satellite energy consumption associated with microservice communication and computation as well as mathematically formulate the energy resource optimization problem for satellite microservices.
- **Augmented Satellite-Microservice Subgraph Construction:** We propose an innovative augmented satellite-microservice subgraph construction method, which equivalently transforms the original problem into a classical DST problem in graph theory, thereby elucidating the mathematical essence of this engineering challenge. In addition to microservice scenarios, this method is also applicable to other routing problems that jointly optimize the computational and communication resource costs.
- **Heuristic Algorithms for the DST Problem:** We implement a series of heuristic algorithms to approximate the optimal solution for the NP-hard DST problem. Extensive experiments demonstrate that our proposed algorithm not only significantly outperforms baseline algorithms in energy efficiency but also exhibits remarkable stability, thereby validating the efficacy of our augmented graph construction methodology.

The remainder of this paper is organized as follows. Section II reviews the related works. Section III presents the system models. The problem formulation is introduced in Section ??.

Section VI details the proposed heuristic algorithms. Section VII shows and analyses the experiment results. Section VIII draws the conclusion.

II. RELATED WORK

Recent advancements in satellite technology have significantly promoted its computing and transmitting capabilities, enabling satellites to undertake some elementary tasks [8]. The construction of ISLs, in particular, enables interconnected satellites to form the SCPN with pooled computing resources, supporting intricate missions with enhanced performance demands [13], [21], [31] and facilitating a global response to requests from anywhere at any time [32], [33]. In addition, careful energy management is necessary for satellites to realize sustainable development [22], [23].

Considering the discrete distributed computing nodes (as satellites) with limited computing power in the SCPN, researchers have migrated the monolithic applications (such as 3D reconstruction and inference model), which are traditionally deployed on ground data centers, to space networks by decomposing them into microservices, which are characterized by being independent, scalable, reusable, and stateless [16], [18], [34]. These microservices can be scaled and deployed on the satellites through platform-independent containerization technology, which encapsulates these isolated microservices into containers that are transparent to the installed computing platform on satellites [14], [35].

Extensive research has been conducted on service routing algorithms, aiming at the problems of node selection and routine planning. Fu et al. [28] proposed a dynamic programming approach to drive the optimal policy on energy allocation for a single satellite in response to user requests. It fails to consider the multi-satellite cases. The work [36] designed a joint edge server placement and service deployment model for SAGIN, aiming to minimize the average response latency and energy cost. The work [22] developed a series of heuristic algorithms to prolong the serving life of satellites by switching part of routers installed on satellites to sleep mode, and resolving the energy-efficient routing policies. Efficient but shortsighted, these proposed algorithms cannot adapt to diverse routing needs, for example, the treed topology raised by requested missions, which demands integrated computing and communication routing as well as multi-satellite collaboration.

III. SYSTEM MODEL

We consider a space computing power network (SCPN) composed of a Walker-Delta LEO constellation, where each satellite is equipped with onboard computing resources and laser inter-satellite links (ISLs). User tasks are represented as ordered microservice chains. Each microservice can have multiple stateless replicas deployed on different satellites, and a request may therefore be executed collaboratively by multiple satellites.

A. LEO Constellation and Time-slotted Topology

A typical Walker-Delta constellation is parameterized by $\mathcal{W} = (A/P/G, H, I)$, where A is the total number of satellites, P is the number of orbital planes, G is the inter-plane phasing factor, H is the orbital altitude, and I is the inclination. Consequently, each orbital plane accommodates $N = A/P$ satellites. The set of satellite nodes in the constellation is formally defined as $\mathcal{V} = \{v_{p,n} \mid p = 1, \dots, P, n = 1, \dots, N\}$, where p denotes the orbital plane index, n denotes the intra-plane position index. To capture the high dynamism of the satellite network, we discretize the constellation's operational period into a set of time slots, as $\mathcal{T} = \{0, 1, \dots, T-1\}$, with a uniform slot duration Δ . For any continuous-time instant τ , the corresponding time slot index is mapped as, $s(\tau) = \lfloor \tau/\Delta \rfloor \bmod T$. Adopting a snapshot-based modeling approach, we assume that the network state, including satellite positions, ISL availability and capacity, remains quasi-static within a single time slot, and only evolves at the slot boundaries governed by orbital mechanics. Thus, the time-varying network topology during slot $t \in \mathcal{T}$ is formally defined as a dynamic and time-dependent graph $\mathcal{G}(t) = (\mathcal{V}, \mathcal{E}(t))$, where $\mathcal{E}(t)$ represents the set of feasible ISLs at slot t .

B. ISL Communication Model

We assume that each satellite is equipped with four laser optical terminals: two positioned along the roll axis for intra-plane ISLs, and two along the pitch axis for cross-plane ISLs.

a) ISL Connectivity Constraints: Each satellite $v_{p,n}$ maintains two intra-plane ISLs with its adjacent neighbors $v_{p,n-}$ and $v_{p,n+}$, where $n^\pm = ((n \pm 1 - 1) \bmod N) + 1$. Because satellites in the same plane share identical angular velocities, their relative distances remain time-invariant, yielding highly stable communication links.

Conversely, cross-plane ISLs are inherently dynamic. Constrained by the physical and kinematic limits of onboard laser communication terminals (LCTs), the existence of a cross-plane link $e = (u, v)$ at slot t is jointly dictated by line-of-sight (LoS) occlusion, mechanical tracking thresholds, and polar exclusions, formulated as:

$$e \in \mathcal{E}(t) \iff \begin{cases} d_{u,v}(t) \leq d_{\max}, \\ \theta_{u,v}(t) \leq \theta_{\max}, \\ \max(|\varphi_u(t)|, |\varphi_v(t)|) \leq \varphi_{\max}. \end{cases} \quad (1)$$

In (32), the LoS distance $d_{u,v}(t)$ and relative tracking angle $\theta_{u,v}(t)$ are bounded by the terrestrial occlusion limit $d_{\max}$ and the hardware threshold $\theta_{\max}$, respectively. Furthermore, the third constraint enforces a polar exclusion zone at latitude boundary $\varphi_{\max}$ (e.g., 70°) through proactively tearing down the cross-plane links at high latitudes, which effectively circumvents the severe pointing dynamics and extreme Doppler shifts inherent to polar regions.

b) ISL Capacity Model: Unlike ground networks with stable capacities, the transmission rate of laser ISLs is distance-dependent due to geometric propagation loss. Based on the Free-Space Optical (FSO) communication model, the

received optical power $P_e^{\text{rx}}(t)$ at satellite v transmitted from satellite u is formalized as:

$$P_e^{\text{rx}}(t) = P^{\text{tx}} G^{\text{tx}} G^{\text{rx}} \eta_{\text{tx}} \eta_{\text{rx}} \left(\frac{\lambda}{4\pi d_{u,v}(t)} \right)^2, \quad (2)$$

where P^{tx} is the nominal laser transmit power, G^{tx} and G^{rx} represent the optical gains of the transmitter and receiver apertures respectively, λ is the laser wavelength, and $\eta_{\text{tx}}, \eta_{\text{rx}}$ denote the optical efficiency factors. The real-time Signal-to-Noise Ratio (SNR) of the link e at time slot t is given as:

$$\gamma_e(t) = \frac{(R_{\text{pd}} \cdot P_e^{\text{rx}}(t))^2}{\sigma_{\text{sh}}^2 + \sigma_{\text{th}}^2}, \quad (3)$$

where R_{pd} is the photodiode responsivity, while σ_{sh}^2 and σ_{th}^2 denote the shot and thermal noise variances, respectively. According to the Shannon-Hartley theorem, the achievable transmission capacity $R_e^0(t)$ is constrained by:

$$R_e^0(t) = B_e \log_2(1 + \gamma_e(t)), \quad (4)$$

where B_e is the modulation bandwidth.

c) ISL Delay Model: The nominal rate $R_e^0(t)$ in (36) could be degraded in practice by transient channel fading and traffic contention. To capture these factors, we introduce a scaling factor $\eta_e(t) \in (0, 1]$ to formulate the effective data rate as $R_e(t) = \eta_e(t) R_e^0(t)$. When transmitting a data block of size B over link $e = (u, v)$ during slot t , the aggregate link-level communication delay comprises queuing, transmission, and propagation delays, formulated as:

$$T_e^{\text{cm}}(B, t) = \tau_e^{\text{qe}}(t) + \tau_e^{\text{tr}}(t) + \tau_e^{\text{pr}}(t). \quad (5)$$

Here, $\tau_e^{\text{qe}}(t)$ denotes the queuing delay, which can be dynamically estimated via lightweight queue-length probing. The transmission delay is given by $\tau_e^{\text{tr}}(t) = B/R_e(t)$, and the propagation delay is $\tau_e^{\text{pr}}(t) = d_e(t)/c$, with c being the speed of light in vacuum.

C. Onboard Computing Model

Let F_v^0 denote the nominal computing capacity of satellite v . In practice, the available computational resources are often constrained by background payload tasks, operating system overhead, and thermal throttling. To capture these hardware and software limitations, we introduce a computational efficiency factor $\xi_v(t) \in (0, 1]$ and define the effective dynamic computing capacity as $F_v(t) = \xi_v(t) F_v^0$.

Let $\tau_v^{\text{qe}}(t)$ denote the queuing delay before scheduling microservice m is executed on satellite v , which increases with the pending task queue length at slot t . The total computation delay is modeled as:

$$T_m^{\text{cp}}(v, t) = \tau_v^{\text{qe}}(t) + \tau_v^{\text{cp}}(t), \quad (6)$$

where $\tau_v^{\text{cp}}(t) = C_m/F_v(t)$ represents the actual execution delay, and C_m is the number of CPU cycles required by microservice m . Correspondingly, the computational energy consumption is defined as:

$$E_m^{\text{cp}}(v, t) = P_v^{\text{cp}} \frac{C_m}{F_v(t)}, \quad (7)$$

where P_v^{cp} is the operational computing power of the satellite.

D. LEO Microservice and Service Request Models

To facilitate flexible application management within the SCPN, complex applications are decoupled into a set of fine-grained, loosely coupled microservices, denoted by $\mathcal{M} = \{1, 2, \dots, M\}$. Each microservice $m \in \mathcal{M}$ is characterized by its computational demand C_m (in CPU cycles), container image size I_m , and storage requirement S_m .

We define a binary indication variable $x_{m,v}(t) \in \{0, 1\}$, which equals to 1 if m is deployed on v during slot t . Accordingly, the subset of satellite nodes hosting a valid replica of microservice m at slot t is denoted as $\mathcal{R}_m(t) = \{v \in \mathcal{V} \mid x_{m,v}(t) = 1\}$. To ensure service availability while preventing excessive resource contention, the total number of deployed replicas for any microservice should be bounded within predefined thresholds, $r_m^{\min}$ and $r_m^{\max}$:

$$r_m^{\min} \leq \sum_{v \in \mathcal{V}} x_{m,v}(t) \leq r_m^{\max}, \quad \forall m \in \mathcal{M}. \quad (8)$$

Furthermore, the aggregate storage footprint of all replicas hosted on a single satellite must not exceed its onboard storage capacity $S_v^{\max}$:

$$\sum_{m \in \mathcal{M}} x_{m,v}(t) S_m \leq S_v^{\max}, \quad \forall v \in \mathcal{V}. \quad (9)$$

It is worth noting that all microservices operate in a stateless manner, which can be dynamically added, removed, or migrated without incurring the prohibitive communication overhead of inter-replica state synchronization.

Let $r = \langle v_s, m_1, m_2, \dots, m_L, v_d, \tau_{\text{in}} \rangle$ denote an incoming request with a deadline D_r . Here, v_s and v_d are the source and destination satellites, m_i is the i -th microservice in the sequential execution chain, and τ_{in} is the continuous arrival time. Let $B_{r,i}$ be the data volume that should be transmitted before invoking m_i , while $B_{r,L+1}$ denotes the final output routed to v_d . Since a request can arrive at any arbitrary instant and its data transmission may span multiple topology slots, ELARA formulates this sequential service execution as a continuous-time online decision process. The detailed cross-slot execution dynamics, decision variables, and objective functions are formally presented in Section V.

IV. PROBLEM FORMULATION

To find a energy- and latency-efficient service routing path for user request in time-varying SCPN environment, we formulate the service routing problem as a online optimization problem, comprising of computation node (satellite) selection, data routing, and service redeployment.

A. Cross-slot Execution and Decision Variables

ELARA tracks the execution of each microservice chain continuously in the temporal domain. Prior to executing microservice m_i , the request state is defined as $\sigma_{r,i} = (u_{r,i}, \tau_{r,i}, i)$, where $u_{r,i}$ is the satellite holding the intermediate input data and $\tau_{r,i}$ is the decision epoch.

To formalize replica selection, we introduce a binary decision variable $z_{r,i,v} \in \{0, 1\}$, which equals 1 if m_i of request

r is executed on satellite v . Each microservice stage must be processed by exactly one valid replica:

$$\sum_{v \in \mathcal{V}} z_{r,i,v} = 1, \quad \text{and} \quad z_{r,i,v} \leq x_{m_i,v}(s(\tau_{r,i})), \quad \forall r, i, v. \quad (10)$$

Let $v_{r,i}$ denote the selected execution satellite, as $z_{r,i,v_{r,i}} = 1$.

For data transmission, let $f_{r,i,e}(t) \geq 0$ denote the data volume carried by link e in slot t for routing segment i . This segment routes intermediate data $B_{r,i}$ from $u_{r,i}$ to $v_{r,i}$ (for $i \leq L$), and the final output $B_{r,L+1}$ from $v_{r,L}$ to v_d . Accumulating the cross-slot routing delay $T_{r,i}^{\text{ru}}$ and energy $E_{r,i}^{\text{ru}}$, the data arrival time and computation finish time at stage i are:

$$\tau_{r,i}^{\text{arr}} = \tau_{r,i} + T_{r,i}^{\text{ru}}, \quad (11)$$

$$\tau_{r,i}^{\text{fin}} = \tau_{r,i}^{\text{arr}} + T_{m_i}^{\text{cp}}(v_{r,i}, s(\tau_{r,i}^{\text{arr}})). \quad (12)$$

The request state then advances via $u_{r,i+1} = v_{r,i}$ and $\tau_{r,i+1} = \tau_{r,i}^{\text{fin}}$. Upon completion of m_L , the end-to-end latency is simply $T_r^{\text{e2e}} = \tau_r^{\text{out}} - \tau_{\text{in}}$, where τ_r^{out} is the final destination arrival time.

Concurrently, the slow-layer redeployment action in slot t is defined as $a_t \in \{\text{add}, \text{remove}, \text{move}, \text{no-op}\}$. These actions are strictly constrained by the storage and replica-count bounds defined in (10) and (11).

B. Latency, Energy, and Migration Cost

Following the delay decomposition in (5), the unit delay and unit energy costs for transmitting data over link e in slot t are:

$$c_e^T(t) = \frac{1}{R_e(t)} + \tau_e^{\text{pr}}(t) + \tau_e^{\text{qe}}(t) + \tau^{\text{sw}}, \quad (13)$$

$$c_e^E(t) = \frac{P_e^{\text{tx}}}{R_e(t)}, \quad (14)$$

where τ^{sw} represents the per-hop switching overhead. Incorporating $\phi_e(t)$ as the short-horizon disruption risk of link e , the composite edge cost driving the routing layer is:

$$c_e(t) = \omega_T c_e^T(t) + \omega_E c_e^E(t) + \omega_R \phi_e(t). \quad (15)$$

For a complete request r , the aggregate routing costs (T_r^{ru} , E_r^{ru}) and computing costs (T_r^{cp} , E_r^{cp}) are formulated by summing the respective metrics across all L stages and the final delivery segment. The total request energy is $E_r^{\text{tot}} = E_r^{\text{ru}} + E_r^{\text{cp}}$. Furthermore, the network-wide migration cost induced by action a_t is normalized as:

$$C^{\text{mig}}(a_t) = \omega_T^{\text{mig}} \frac{T^{\text{mig}}(a_t)}{T_0} + \omega_E^{\text{mig}} \frac{E^{\text{mig}}(a_t)}{E_0}, \quad (16)$$

with T^{mig} and E^{mig} representing the overhead of transferring service images (evaluating to zero for remove and no-op actions).

C. Optimization Objective

Let N_r^{cro} denote the number of slot crossings experienced by request r , and $\mathbb{I}_r^{\text{fail}} \in \{0, 1\}$ indicate a failure to meet deadline D_r . The normalized penalty function for request r is:

$$J_r = \omega_T \frac{T_r^{\text{e2e}}}{T_0} + \omega_E \frac{E_r^{\text{tot}}}{E_0} + \omega_S N_r^{\text{cro}} + \omega_F \mathbb{I}_r^{\text{fail}}. \quad (17)$$

Given a sequence of requests $\mathcal{Q}$, the comprehensive online optimization problem is formulated as:

$$\mathbf{P1} : \min_{\mathbf{z}, \mathbf{f}, \mathbf{a}} \sum_{r \in \mathcal{Q}} J_r + \omega_M \sum_{t \in \mathcal{T}} C^{\text{mig}}(a_t) \quad (18a)$$

$$\text{s.t.} \quad (10), (11), (15), \quad (18b)$$

$$0 \leq f_{r,i,e}(t) \leq R_e(t) \Delta_{r,i}^{\text{rem}}(t), \quad \forall r, i, e, t, \quad (18c)$$

$$\sum_{e \in \delta^+(u)} f_{r,i,e}(t) - \sum_{e \in \delta^-(u)} f_{r,i,e}(t) = b_{r,i,u}(t), \quad \forall r, i, u, t, \quad (18d)$$

$$a_t \in \{\text{add}, \text{remove}, \text{move}, \text{no-op}\}, \quad \forall t, \quad (18e)$$

$$T_r^{\text{e2e}} \leq D_r \quad \text{or} \quad \mathbb{I}_r^{\text{fail}} = 1, \quad \forall r. \quad (18f)$$

Constraint (31d) enforces flow conservation across the dynamic graph, where $b_{r,i,u}(t)$ defines the net data balance: positive at the source, negative at the selected destination replica, and zero at intermediate relay satellites.

D. Problem Hardness

Proposition 1. Problem **P1** is strictly NP-hard.

Proof Sketch. The intractable nature of **P1** can be verified by analyzing a restricted static snapshot of the system. Specifically: 1) even with a single time slot and fixed topology, selecting replicas for a microservice chain reduces to the classical NP-hard Service Function Chain (SFC) placement problem; 2) connecting these dispersed replicas subject to link capacities entails constrained shortest-path routing, which is inherently NP-hard; and 3) the full dynamic formulation introduces cross-slot routing, time-varying queues, and adaptive redeployment, strictly amplifying the decision space complexity. Consequently, solving **P1** exactly via offline optimization is intractable for continuous Space Computing Power Network (SCPN) operations, necessitating ELARA's architectural decomposition into three coupled, scalable online agents. ■

V. PROBLEM FORMULATION

The proposed ELARA is designed to jointly determines which replica executes each microservice, how data is routed across time-varying ISLs, and when service replicas should be added, removed, or moved. The formulation captures the ELARA design without assuming full knowledge of future requests, future queue states, or future link-rate fluctuations.

A. Decision Variables

For request r , stage i , and satellite v , the fast-layer execution decision is

$$z_{r,i,v} = \begin{cases} 1, & \text{if } m_i \text{ of } r \text{ is executed on } v, \\ 0, & \text{otherwise.} \end{cases} \quad (19)$$

Each stage must be executed by exactly one available replica:

$$\sum_{v \in \mathcal{V}} z_{r,i,v} = 1, \quad \forall r, i, \quad (20)$$

and

$$z_{r,i,v} \leq x_{m_i,v}(s(\tau_{r,i})), \quad \forall r, i, v. \quad (21)$$

For routing hop i of request r , let $f_{r,i,e}(t) \geq 0$ be the amount of data transmitted through link e in slot t . Its capacity constraint is

$$0 \leq f_{r,i,e}(t) \leq R_e(t) \Delta_{r,i}^{\text{rem}}(t), \quad \forall e \in \mathcal{E}(t), \quad (22)$$

where $\Delta_{r,i}^{\text{rem}}(t)$ is the remaining usable time in slot t . At each slot boundary, the residual data is updated and the routing decision is recomputed.

The slow-layer redeployment action in slot t is

$$a_t \in \{\text{add}, \text{remove}, \text{move}, \text{no-op}\}. \quad (23)$$

Every action must preserve (10) and (11). For add and move actions, the target satellite must have enough residual storage and must not already host the same microservice. For remove actions, the service must still have at least r_m^{min} replicas after removal.

B. Latency, Energy, and Migration Cost

For one unit of data on link e in slot t , the unit delay cost follows the communication-delay decomposition in (5):

$$c_e^T(t) = \frac{1}{R_e(t)} + \tau_e^{\text{pr}}(t) + \tau_e^{\text{qe}}(t) + \tau^{\text{sw}}, \quad (24)$$

where τ^{sw} denotes per-hop switching overhead. The unit communication energy is

$$c_e^E(t) = P_e^{\text{tx}} \frac{1}{R_e(t)}. \quad (25)$$

Let $\phi_e(t)$ be the short-horizon link disruption risk estimated from future ISL availability over a small look-ahead window. The routing-layer edge cost is

$$c_e(t) = \omega_T c_e^T(t) + \omega_E c_e^E(t) + \omega_R \phi_e(t). \quad (26)$$

For request r , the total communication delay and energy are

$$T_r^{\text{ru}} = \sum_{i=1}^{L+1} T_{r,i}^{\text{ru}}, \quad E_r^{\text{ru}} = \sum_{i=1}^{L+1} E_{r,i}^{\text{ru}}. \quad (27)$$

The computation delay and energy are

$$T_r^{\text{cp}} = \sum_{i=1}^L \sum_{v \in \mathcal{V}} z_{r,i,v} T_{m_i}^{\text{cp}}(v, s(\tau_{r,i}^{\text{arr}})), \quad (28)$$

and

$$E_r^{\text{cp}} = \sum_{i=1}^L \sum_{v \in \mathcal{V}} z_{r,i,v} E_{m_i}^{\text{cp}}(v, s(\tau_{r,i}^{\text{arr}})). \quad (29)$$

Thus,

$$E_r^{\text{tot}} = E_r^{\text{ru}} + E_r^{\text{cp}}. \quad (30)$$

For a replica redeployment action a_t , the migration cost is

$$C^{\text{mig}}(a_t) = \omega_T^{\text{mig}} \frac{T^{\text{mig}}(a_t)}{T_0} + \omega_E^{\text{mig}} \frac{E^{\text{mig}}(a_t)}{E_0}, \quad (31)$$

where $T^{\text{mig}}(a_t)$ and $E^{\text{mig}}(a_t)$ are the delay and energy of transferring the service image. Remove and no-op actions have zero migration-transfer cost.

C. Optimization Objective

Let N_r^{cro} be the number of slot crossings experienced by request r , and let $\mathbb{I}_r^{\text{fail}}$ indicate whether the request fails. The per-request cost is

$$J_r = \omega_T \frac{T_r^{\text{e2e}}}{T_0} + \omega_E \frac{E_r^{\text{tot}}}{E_0} + \omega_S N_r^{\text{cro}} + \omega_F \mathbb{I}_r^{\text{fail}}. \quad (32)$$

Given a sequence of requests $\mathcal{Q}$, the online problem is

$$\mathbf{P1}: \min_{\mathbf{z}, \mathbf{f}, \mathbf{a}} \sum_{r \in \mathcal{Q}} J_r + \omega_M \sum_{t \in \mathcal{T}} C^{\text{mig}}(a_t) \quad (33a)$$

$$\text{s.t. (14), (15),} \quad (33b)$$

$$0 \leq f_{r,i,e}(t) \leq R_e(t) \Delta_{r,i}^{\text{rem}}(t), \quad \forall r, i, e, t, \quad (33c)$$

$$\sum_{e \in \delta^+(u)} f_{r,i,e}(t) - \sum_{e \in \delta^-(u)} f_{r,i,e}(t) = b_{r,i,u}(t), \quad \forall r, i, u, t, \quad (33d)$$

$$(10), (11), \quad (33e)$$

$$a_t \in \{\text{add, remove, move, no-op}\}, \quad \forall t, \quad (33f)$$

$$T_r^{\text{e2e}} \leq D_r \quad \text{or} \quad \mathbb{I}_r^{\text{fail}} = 1, \quad \forall r. \quad (33g)$$

In (31d), $b_{r,i,u}(t)$ is the net flow balance for routing hop i of request r : it is positive at the current source, negative at the selected execution or destination node, and zero at relay satellites. The formulation is online because the controller observes the current topology, deployment state, link queues, and compute queues, while future requests and future queue dynamics are unknown.

D. Problem Hardness

Proposition 1. Problem P1 is NP-hard.

Sketch. Consider a restricted case with a single time slot, fixed replica deployment, fixed effective link rates and compute capacities, and no redeployment. The controller still needs to select one replica for each service in a chain and connect consecutive selected replicas with feasible paths to minimize joint computation and communication cost. This restricted case contains the classical service placement and routing problem as a special case and can encode constrained shortest-path and facility-location-like decisions. Therefore, the restricted problem is NP-hard, and the full problem with cross-slot

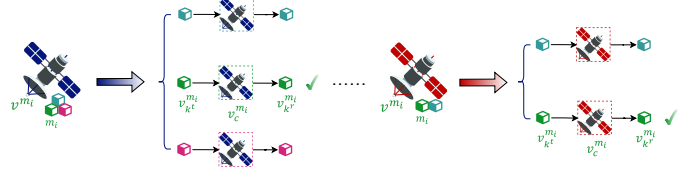


Fig. 2: Abstract the computation process from colocated microservice and satellite deployment.

routing, stochastic load, and adaptive redeployment is at least as hard.

Solving P1 exactly is not suitable for online SCPN operation because requests arrive continuously, topology changes every slot, and redeployment decisions have delayed effects. ELARA therefore decomposes the problem into three coupled online layers: a fast PPO-GNN replica selector, a cross-slot min-cost max-flow router, and a slow bandit-based replica redeployment agent.

VI. ENERGY-AWARE SERVICE ROUTING ALGORITHM FOR MICROSERVICES IN THE SCPN

In this section, we present our approach to addressing the proposed problem P as outlined in Section IV. We first detail our augmented satellite-microservice connected sub-graph construction algorithm, and then elaborate on the design of our DST-based service routing algorithm tailored to optimize energy consumption in the SCPN.

A. Augmented Satellite-Microservice Sub-Graph Construction

For each request originating from a terrestrial station, there is a DAG $\mathcal{G}_f^m$, which outlines the service execution sequence. In $\mathcal{G}_f^m$, there may exist ρ ($1 \leq \rho \leq |\mathcal{M}_f| - 1$) vertices whose outdegree equals 0. These vertices represent terminal microservices that conclude the processing sequence by dispatching results to their respective destinations. Consequently, there should be multiple logical destination satellite nodes, denoted as $\{v^{dst_1}, \dots, v^{dst_\rho}\}$. The routing paths from the logical destination satellites hosting the terminal microservices to the real destination satellites can be straightforwardly generated by the control panel. Note that if a terminal microservice is deployed directly on its real destination satellite, this satellite also acts as the logical destination. In such cases, the communication cost between the terminal service and its destination is effectively zero. Next, we define the augmented graph to be constructed as $\mathcal{G}_z = (\mathcal{V}_z, \mathcal{E}_z)$, where initially, $\mathcal{V}_z = \emptyset$ and $\mathcal{E}_z = \emptyset$. Then, the process of constructing graph $\mathcal{G}_z$ is detailed as follows:

Step 1. This step involves adding the source node v^{src} , along with the logical destination nodes $\{v^{dst_1}, \dots, v^{dst_\rho}\}$ and the real destination node v_{dst} , to vertex collection $\mathcal{V}_z$.

Step 2. For each microservice m_i in $\mathcal{M}_f$, and $\forall v_k \in \mathcal{V}_{m_i}$, this step creates three distinct vertices: $v_{k^r}^{m_i}$, $v_{k^c}^{m_i}$ and $v_{k^t}^{m_i}$, which are then added to the set $\mathcal{V}_z$. Subsequently, it constructs two directed edges $v_{k^r}^{m_i} \rightarrow v_{k^c}^{m_i}$ and $v_{k^c}^{m_i} \rightarrow v_{k^t}^{m_i}$, both with weights equal to $E_{m_i}(v_k)/2$. These edges are added to the edge set $\mathcal{E}_z$. This procedure is detailed in Fig. 2.

Algorithm 1: Augmented Satellite-Microservice Connected Subgraph Construction Algorithm.

```

1 Initialize augmented directed graph  $\mathcal{G}_z$  with vertex set
   $\mathcal{V}_z \leftarrow \emptyset$  and edge set  $\mathcal{E}_z \leftarrow \emptyset$ ;
2  $\mathcal{V}_z \leftarrow \mathcal{V}_z \cup \{v^{src}, v^{dst_1}, \dots, v^{dst_\rho}\}$ ;
3 for  $m_i$  in  $\mathcal{M}_f$  do
4   for  $v_k$  in  $\mathcal{V}_{m_i}$  do
5      $\mathcal{V}_z \leftarrow \mathcal{V}_z \cup \{v_{k^r}^{m_i}, v_{k^c}^{m_i}, v_{k^t}^{m_i}\}$ ;
6      $\mathcal{E}_z \leftarrow \mathcal{E}_z \cup \{(v_{k^r}^{m_i} \rightarrow v_{k^c}^{m_i}), (v_{k^c}^{m_i} \rightarrow v_{k^t}^{m_i})\}$ ;
7     Set  $w(v_{k^r}^{m_i} \rightarrow v_{k^c}^{m_i}) \doteq E_{m_i}(v_k)/2$ ;
8     Set  $w(v_{k^c}^{m_i} \rightarrow v_{k^t}^{m_i}) \doteq E_{m_i}(v_k)/2$ ;
9   end
10 end
11 for  $w_{i,j}$  in  $\mathcal{W}_f$  do
12   for  $v_k^{m_i}$  in  $\mathcal{V}_{m_i}$  do
13     for  $v_{k'}^{m_j}$  in  $\mathcal{V}_{m_j}$  do
14        $\mathcal{V}_o^{re}, \mathcal{E}_o^{re} = \text{GISRSA}(\mathcal{G}, v_k^{m_i}, v_{k'}^{m_j})$ ;
15        $\mathcal{V}_z \leftarrow \mathcal{V}_z \cup \mathcal{V}_o^{re}$ ;
16        $\mathcal{E}_z \leftarrow \mathcal{E}_z \cup \mathcal{E}_o^{re}$ ;
17     end
18   end
19 end
20 for  $v_k^{m_{src}}$  in  $\mathcal{V}_{m_{src}}$  do
21    $\mathcal{V}_o^{re}, \mathcal{E}_o^{re} = \text{GISRSA}(\mathcal{G}, v^{src}, v_k^{m_{src}})$ ;
22    $\mathcal{V}_z \leftarrow \mathcal{V}_z \cup \mathcal{V}_o^{re}$ ;
23    $\mathcal{E}_z \leftarrow \mathcal{E}_z \cup \mathcal{E}_o^{re}$ ;
24 end
25 for  $v_k^{m_{dst}}$  in  $\mathcal{V}_{m_{dst}}$  do
26   for  $g$  in  $[1, \rho]$  do
27      $\mathcal{V}_o^{re}, \mathcal{E}_o^{re} = \text{GISRSA}(\mathcal{G}, v_k^{m_{dst}}, v^{dst_g})$ ;
28      $\mathcal{V}_z \leftarrow \mathcal{V}_z \cup \mathcal{V}_o^{re}$ ;
29      $\mathcal{E}_z \leftarrow \mathcal{E}_z \cup \mathcal{E}_o^{re}$ ;
30   end
31 end

```

Step ③. For the calling relationship between two microservices m_i and m_j , that is, $\forall w_{i,j} \in \mathcal{W}_f$, the algorithm will iteratively identify all satellite nodes from collection $\mathcal{V}$ that host microservices m_i and m_j , respectively. If satellites $v_k^{m_i}$ and $v_{k'}^{m_j}$ are directly adjacent without needing any relay satellite to facilitate the communication, then a directed edge $v_{k^r}^{m_i} \rightarrow v_{k^t}^{m_j}$ is constructed, and its weight is set as the communication energy cost from $v_{k^r}^{m_i}$ to $v_{k^t}^{m_j}$ in $\mathcal{G}$, given as:

$$e^{tr} = P_T^{v_{k^r}^{m_i} \rightarrow v_{k^t}^{m_j}} \frac{w_{i,j}}{w(v_{k^r}^{m_i} \rightarrow v_{k^t}^{m_j})}, \quad (34)$$

Otherwise, the algorithm then searches for available relay satellite nodes within $\mathcal{V}$, which can be efficiently resolved using the Floyd algorithm, resulting in a collection of relay satellite nodes denoted as $\mathcal{V}^{re} = \{v_1^{re}, \dots, v_\xi^{re}\}$. Then, it adds these relay nodes to $\mathcal{V}_z$, and constructs and adds edges $v_{k^r}^{m_i} \rightarrow v_a^{re}, v_a^{re} \rightarrow v_{a+1}^{re}, v_\xi^{re} \rightarrow v_{k^r}^{m_j}, (\forall a \in [1, \xi - 1])$ to $\mathcal{E}_z$. The weights of these edges, representing the relay

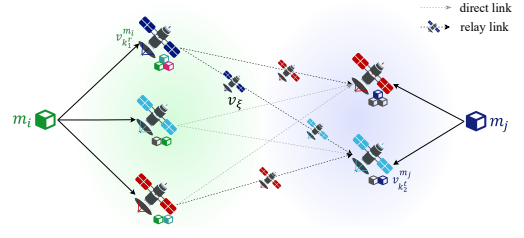


Fig. 3: Build the relay routing paths for neighbor microservices that are deployed on non-adjacent satellites.

communication energy cost between satellites, are calculated according to Eq. ?? and ??, as below:

$$e_{re1}^{tr}(v_{k^t}^{m_i} \rightarrow v_a^{re}) = P_T^{v_{k^t}^{m_i} \rightarrow v_a^{re}} \frac{w_{i,j}}{w(v_{k^t}^{m_i} \rightarrow v_a^{re})}, \quad (35)$$

$$e_{re2}^{tr}(v_a^{re} \rightarrow v_{a+1}^{re}) = P_T^{v_a^{re} \rightarrow v_{a+1}^{re}} \frac{w_{i,j}}{w(v_a^{re} \rightarrow v_{a+1}^{re})}, \quad (36)$$

$$e_{re3}^{tr}(v_\xi^{re} \rightarrow v_{k^r}^{m_j}) = P_T^{v_\xi^{re} \rightarrow v_{k^r}^{m_j}} \frac{w_{i,j}}{w(v_\xi^{re} \rightarrow v_{k^r}^{m_j})}. \quad (37)$$

This process is depicted in Fig. 3.

Step ④. For the first executed microservice m_{src} and the final executed microservices $\{m_{dst^1}, \dots, m_{dst^\rho}\}$, the algorithm searches for the collection of satellite nodes hosting these microservices. Subsequently, it constructs the edges $v^{src} \rightarrow v_{k^r}^{m_{src}}$ and $v_{k^t}^{m_{dst}} \rightarrow v^{dst_g}, \forall g \in [1, \rho]$. The detailed procedures for the construction of the newly generated vertices and edges are similar to those outlined in **Step ③**.

We summarize the above steps in Algorithm 1. This systematic approach guarantees that within the augmented graph $\mathcal{G}_z$, each tree rooted at the source node v^{src} , which includes the destination node v^{dst} and all $v_{k^c}^{m_i}$ nodes, constitutes a feasible service routing scheme ψ . This scheme ensures that all computation and communication processes required in $\mathcal{G}_{f'}^m$ are executed sequentially, and the sum of the weights of all edges in this tree quantifies the total energy E^{total} required for the routing scheme. Subsequently, solving the original problem can essentially be transformed into tackling a DST problem on the graph $\mathcal{G}_z$. The objective is to identify a connected subgraph that contains all specified nodes in graph $\mathcal{G}_z$ such that the sum of the weights of all edges in the subgraph is minimized. The directed Steiner tree problem is defined as follows.

Definition 4. (Directed Steiner Tree). *Given a directed graph $G=(V, A)$, an assigned root node $r \in V$, and a set of terminals $X \subseteq V, (|X| = k)$, the objective is to find the minimum cost arborescence rooted at r and spanning all the vertices in X .*

B. DST-based Service Routing Algorithm

The DST problem is known to be NP-hard, and extensive research has been conducted on it. Numerous studies have proposed approximation solutions for the original problem [29], [30]. However, solving the DST problem within the LEO network topology introduces unique and particularly severe challenges due to the dynamic nature of satellite constellations. Specifically, the time-varying nature of ISLs owing to the constant movement of satellites results in a highly dynamic

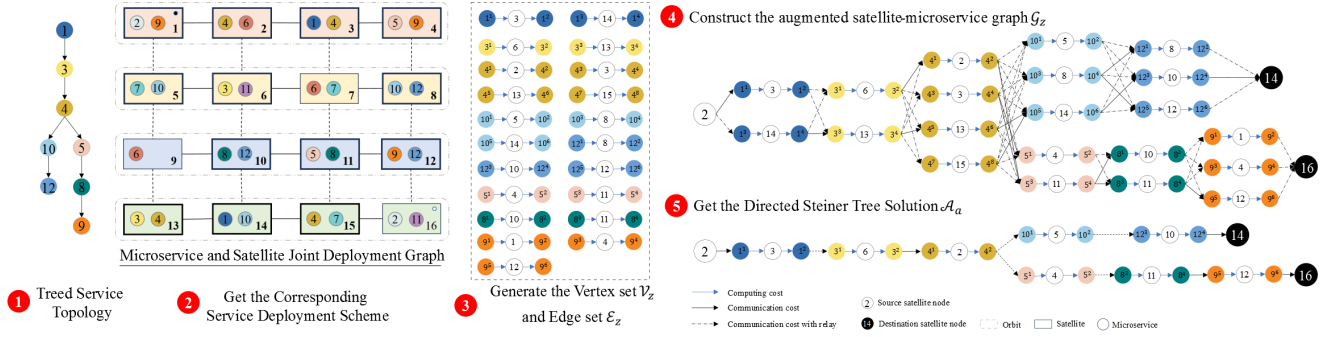


Fig. 4: The overview of solution workflow of tree-based request topology.

Algorithm 2: Approximated Solution for Minimized Spanning Directed Steiner Arborescence.

Input : $\mathcal{G}_z, \{v^{src}, v^{dst_1}, \dots, v^{dst_g}\}$
Output: $\mathcal{A}(\mathcal{V}_a, \mathcal{E}_a)$

- 1 Init the terminal pairs set, $\{(v^{src}, v^{dst_g}) | \forall g \in [1, \rho]\}$;
- 2 Init $\mathcal{V}_a \leftarrow \emptyset, \mathcal{E}_a \leftarrow \emptyset$;
- 3 **for** g in $[1, \rho]$ **do**
- 4 $\mathcal{V}_g, \mathcal{E}_g = \text{Dijkstra}(\mathcal{G}_z, v^{src}, v^{dst_g})$
- 5 **end**
- 6 $\mathcal{V}_a \leftarrow \bigcap_{g=1}^{\rho} \mathcal{V}_g$;
- 7 $\mathcal{E}_a \leftarrow \mathcal{E}_1$;
- 8 **for** g in $[2, \rho]$ **do**
- 9 **for** e in $\mathcal{E}_g$ **do**
- 10 **if** $e \in \mathcal{E}_a$ **then**
- 11 $w(e) \doteq \min(w(\mathcal{E}_a), w(\mathcal{E}_g))$
- 12 **else**
- 13 $\mathcal{E}_a \leftarrow \mathcal{E}_a \cup \{e\}$
- 14 **end**
- 15 **end**
- 16 **end**
- 17 Construct $\mathcal{A}(\mathcal{V}_a, \mathcal{E}_a)$;

Algorithm 3: Generate the Inter-Satellite Relay Routing Scheme Algorithm (GISRA).

Input : $\mathcal{G}, v_k^{m_i}, v_k^{m_j}$
Output: $\mathcal{V}_o^{re}, \mathcal{E}_o^{re}$

- 1 Initialize the vertex set and edge set with $\mathcal{V}_o^{re} \leftarrow \emptyset, \mathcal{E}_o^{re} \leftarrow \emptyset$;
- 2 **if** $v_k^{m_i}$ is adjacent to $v_k^{m_j}$ **then**
- 3 $\mathcal{E}_o^{re} \leftarrow \mathcal{E}_z \cup \{(v_k^{m_i} \rightarrow v_k^{m_j})\}$;
- 4 $w(v_k^{m_i} \rightarrow v_k^{m_j}) \doteq e^{tr}(v_k^{m_i} \rightarrow v_k^{m_j})$;
- 5 **else**
- 6 $\mathcal{V}_o^{re}, \text{path} = \text{Floyd}(\mathcal{G}, v_k^{m_i}, v_k^{m_j})$;
- 7 $\mathcal{V}_o^{re} \leftarrow \mathcal{V}_o^{re} \cup \{v_k^{m_i}, v_k^{m_j}\}$;
- 8 $\mathcal{E}_o^{re} \leftarrow \mathcal{E}_o^{re} \cup \{(v_k^{m_i} \rightarrow v_1^{re}), (v_1^{re} \rightarrow v_k^{m_j})\}$;
- 9 $w(v_k^{m_i} \rightarrow v_1^{re}) \doteq e^{tr}_{re1}(v_k^{m_i} \rightarrow v_1^{re})$;
- 10 $w(v_1^{re} \rightarrow v_k^{m_j}) \doteq e^{tr}_{re3}(v_1^{re} \rightarrow v_k^{m_j})$;
- 11 **for** $a \in [1, \xi - 1]$ **do**
- 12 $\mathcal{V}_o^{re} \leftarrow \mathcal{V}_o^{re} \cup \{v_a^{re}, v_{a1}^{re}\}$;
- 13 $\mathcal{E}_o^{re} \leftarrow \mathcal{E}_o^{re} \cup \{(v_a^{re} \rightarrow v_{a1}^{re})\}$;
- 14 $w(v_a^{re} \rightarrow v_{a1}^{re}) \doteq e^{tr}_{re2}(v_a^{re} \rightarrow v_{a1}^{re})$;
- 15 **end**
- 16 **end**

network topology, making the link connectivity a complex issue. LoS constraints further complicate the ability to guarantee consistent connections between terminals. Additionally, microservices that need to communicate may not be colocated on the same or directly adjacent satellites, necessitating the use of relay satellites to establish communication paths. These challenges motivate us to design a specialized heuristic algorithm tailored to the ever-changing LEO satellite networks.

The main steps of our energy-aware DST-based service routing algorithm are as follows:

1) Augmented Graph Initialization: In time slot s , the augmented graph $\mathcal{G}_z(s) = (\mathcal{V}_z(s), \mathcal{E}_z(s))$ is initialized with weighed edges.

2) Get the Shortest Path: For each root and terminal node pairs $(v^{src}, v^{dst_g}), \forall g \in [1, \rho]$, it recursively finds the minimum cost path by utilizing the Dijkstra algorithm. Record the involved vertices and edges that constitute each path, represented by $\{\mathcal{V}_1(s), \dots, \mathcal{V}_\rho(s)\}$ and $\{\mathcal{E}_1(s), \dots, \mathcal{E}_\rho(s)\}$.

3) Merge the Shortest Paths: Aggregate the vertex sets

from the minimum cost paths and remove any duplicates to obtain $\mathcal{V}_a(s) = \bigcap_{g=1}^{\rho} \mathcal{V}_g(s)$. Then, consolidate these edge sets to $\mathcal{E}_a(s) = \bigcap_{g=1}^{\rho} \mathcal{E}_g(s)$. Note that if any edges share common vertices, select the edge with the smaller weight to ensure the most cost-effective connections.

4) Minimum Spanning Steiner Arborescence: Utilize the aggregated vertex and edge collections to construct the minimum spanning Steiner arborescence for the snapshot s , denoted by $\mathcal{A}(s) = (\mathcal{V}_a(s), \mathcal{E}_a(s))$. This step provides an approximate solution to the DST problem, optimizing the cost while ensuring connectivity among specified nodes.

Algorithm 2 presents a detailed step-by-step procedure for the proposed approach, offering a comprehensive view of its implementation. To enhance understanding, a simplified example is provided in Fig. 4, which visually demonstrates the key steps and logic of the algorithm.

C. Algorithm Complexity Analysis

The complexity of Algorithm 1 and 2 is analyzed as follows.

Complexity of Algorithm 1. The time complexity of Dijkstra algorithm is given by $\mathcal{O}(|\mathcal{E}| + |\mathcal{V}| \log |\mathcal{V}|)$. In the worst case, the algorithm needs to run $|\mathcal{V}^*|$ times, which is the number of destination satellites, and the total time complexity deteriorates to $\mathcal{O}(|\mathcal{V}^*|(|\mathcal{E}| + |\mathcal{V}| \log |\mathcal{V}|))$. This signifies that the complexity is contingent upon the structure of both the service request graph and the space network topology.

Complexity of Algorithm 2. The time complexity of merging these distinct sets into one set is $\mathcal{O}(n)$, where n is the total number of elements across all sets. Therefore, the time complexity of obtaining the minimum DST is inferred as $\mathcal{O}(|\mathcal{V}^*|(|\mathcal{E}| + |\mathcal{V}| \log |\mathcal{V}|) + 2*n + (|\mathcal{V}_z| + |\mathcal{E}_z|))$, where $\mathcal{O}(2*n)$ represents the time complexity of merging the generated vertex and edge sets, and $\mathcal{O}(|\mathcal{V}_z| + |\mathcal{E}_z|)$ is the time complexity of constructing the directed Steiner tree.

VII. PERFORMANCE EVALUATION

A. Experiment Setup

Constellation. In experiments, we consider a 72/12/1 Telesat Star Constellation system, which consists of $P = 6$ polar orbit planes whose inclination is 99.5° , and the orbital altitude is $h = 1000km$ on the ground [38].

Satellite Computing Capability. We suppose LEO satellites are equipped with several onboard processors and storage components. According to [27], we assign the satellite computing capability following a uniform distribution, as $\varphi(v_k) \sim \text{DiscreteUniform}(\{100, 160, 220, 280, 340, 400\})$, while the processor power $\Psi(v_k) = \varphi(v_k) \times 0.01$ in Watts.

ISL Parameters. Similar to settings in [32], [33], [39], the transmission power of all relevant satellites is standardized at $P_T = 3$ (Watt). In the context of the Telesat Constellation, which achieve inter-satellite interconnection across free-space optical link, the carrier frequency of each of satellite is set as $f_c = 193(THz)$, and the corresponding carrier wavelength is $\lambda = 1550(nm)$ while the channel bandwidth between two satellites is defined as $BW = 2\%f_c$. Besides, we specify the antenna diameter transceiver as $D_R = 6(mm)$, the pointing loss

angle $\theta_0 = 0.01\pi$ and the optical conversion efficiency of the transceiver modules $\eta = 0.8$.

Microservices. According to the microservice setting adopted by [27], we heuristically construct the microservices testbed as follows: There are $|\mathcal{M}| = 60$ microservices whose data volume follows a Gaussian distribution with a mean of 50 and variance of 4 in Gbits. For each microservice, there are 4 ~ 6 replicas distributed on satellites, while each satellite can concurrently host at most $\lfloor |\mathcal{M}| \times 6/72 \rfloor$ microservices.

Ablation Solutions: We design the following benchmarks to evaluate the performance of our proposed algorithm.

- 1) **Random:** It randomly selects a satellite for each microservice and uses the Dijkstra algorithm to get the shortest path when dealing with relay links.
- 2) **Computation greedy (Comp-greedy):** It chooses the satellite that offers the minimum computation energy cost for executing each microservice. The relay routing policy is the same as **Random** policy.
- 3) **Communication greedy (Comm-greedy):** It picks the satellite pair with the minimum communication energy cost, including the potential relay links.

We define three service request topologies: the Chained topology, the multi-forked Treed topology, and the Sophisticated Treed (Soph Tree) topology, as depicted in Fig. 5.

To verify the effectiveness of our algorithm, we conducted a series of experiments on a space topology comprising 72 satellites, each hosting 60 microservices with 4~6 replicas. We generated 100 service request chains, varying in length from 10 to 20, to simulate a chained microservice topology. We also constructed 100 random treed topologies, containing 20 to 45, featuring a stochastic number of leaf nodes and varying branch configurations. For each sophisticated treed topology, microservices were randomly assigned to nodes, and this assignment process was independently repeated 100 times. For each topology, energy cost data was collected from 100 samples. The final results were derived by averaging the energy cost values across these samples. It ensures robust and reliable performance evaluation by mitigating variations caused by randomness in the topology generation and microservice allocation processes.

B. Experimental Results

1) **Effectiveness:** As illustrated in Fig. 6(a), our proposed Steiner consistently outperforms all baseline methods across various microservices topology scenarios, and its advantages become more pronounced in more complex topologies. This enhanced performance is charged to our comprehensive approach that simultaneously considers both computation and communication costs when determining the most efficient routing paths. The comp-greedy method outperforms the random method by prioritizing minimizing computation costs. The comm-greedy method surpasses both comp-greedy and random methods by focusing on minimizing communication costs. These findings emphasize the necessity of accounting for both computation and communication factors to design effective and energy-conscious routing strategies for the SCPN.

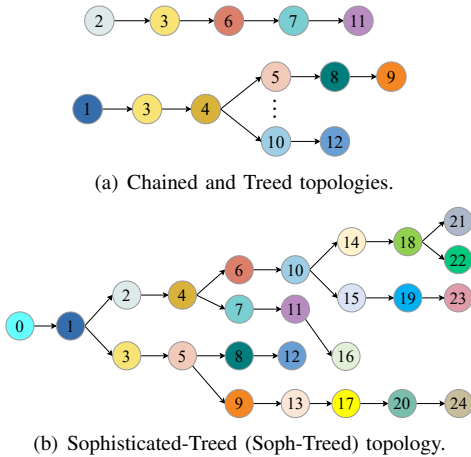


Fig. 5: The overview of service request topologies from terrestrial stations.

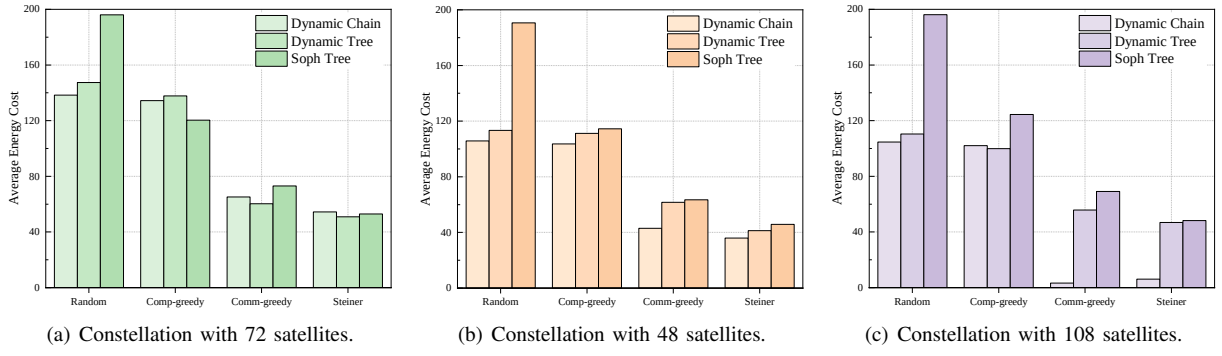


Fig. 6: Performance comparison on energy-cost under varying constellation scales for dynamic microservice request topologies.

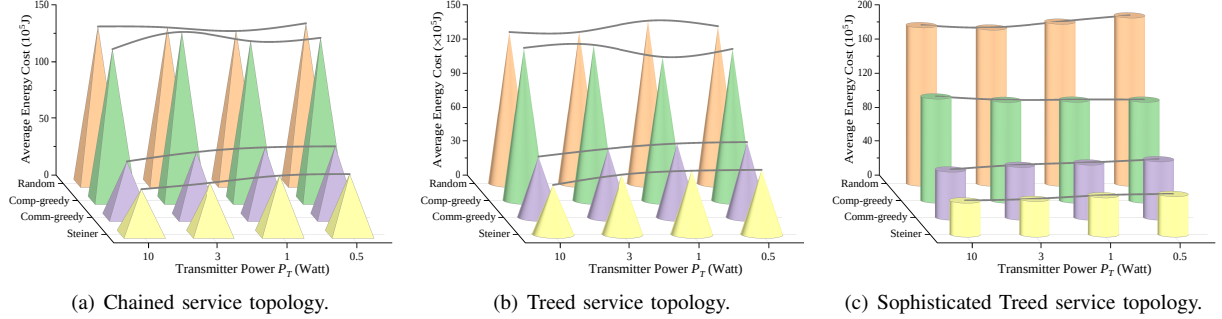


Fig. 7: Performance comparison on energy-cost under different onboard transmitter power configurations for dynamic microservice request topologies.

2) *Impact of constellation scales:* To evaluate the scalability of our algorithm, we systematically varied the number of satellites in the constellation by adjusting the number of orbits while maintaining consistent microservice deployment strategies. Experiments were conducted in two constellation settings separately, as illustrated in Fig. 6(b) and Fig. 6(c). These configurations allowed us to assess the algorithm's performance under varying levels of network complexity and service topology. Our results demonstrate that the proposed approach consistently outperforms alternative methods across all tested constellation scales and topological scenarios. Notably, its effectiveness becomes increasingly pronounced as the network complexity escalates, highlighting its robustness and adaptability to intricate environments.

3) *Impact of transmitter power:* Given the dominant role of communication costs in the total energy consumption of satellite systems, we conducted a series of experiments to explore how varying channel conditions influence the performance of our Steiner scheme. Specifically, we experimented with four different transmitter power levels on satellites across diverse microservice request schemes to examine their effect on energy efficiency. The results shown in Figs. 7(a), 7(b) and 7(c) indicate that as transmitter power increases, the average energy cost shows an upward trend. This trend highlights the trade-off between signal strength and energy consumption, with higher power levels leading to greater energy expenditure. Despite this increase in energy cost, our Steiner algorithm consistently outperformed other benchmark strategies across

all service request scenarios. This superior performance is attributed to the algorithm's ability to dynamically adapt to the complexities of space network topologies and fluctuating channel conditions. By optimizing service routing paths, our approach minimizes unnecessary energy use under varying channel conditions. These findings underscore the importance of intelligent service routing mechanisms in the SCPN, particularly in environments with variable communication links.

4) *Impact of faulty satellites:* Given the constraints of limited power supply and the effects of solar eclipses, some satellites may need to enter a low-power mode, functioning solely as relay nodes for message forwarding [22]. To evaluate the robustness of our proposed method under such conditions, we conducted experiments by randomly disabling a subset of satellites in a constellation of 72 satellites. Specifically, we tested scenarios with 10, 20, and 25 faulty satellites by eliminating their microservices, as shown in Figs 8(a), 8(b), 8(c). The results reveal that energy cost generally increases with an increasing number of faulty satellites for all methods. This trend can be attributed to the reduced availability of active nodes, which forces longer transmission paths and higher energy consumption for maintaining connectivity (much more relay satellites are required for message forwarding). Despite this challenge, Steiner invariably outperforms other methods in all scenarios. Its superior performance is attributed to its robust design, which can effectively adapt to changes in network topology and operational conditions. By optimizing power allocation and service routing decisions, the algorithm

minimizes disruptions caused by satellite failures.

5) *Impact of microservice deployment schemes*: To investigate the influence of microservice deployment and scaling strategies on energy consumption within the SCPN, we conducted experiments by varying the number of replicas allocated for each microservice. Specifically, we tested three replica ranges: [2, 6], [4, 6], and [6, 10]. And the experimental results, illustrated in Figs. 9(a), 9(b), and 9(c), reveal that energy costs generally decrease as the scale of microservice deployment expands. This trend is attributed to the increased availability of service nodes and routing paths, which provide more opportunities for route optimization and efficient power utilization. Our algorithm regularly demonstrates superior performance in reducing energy costs compared to other benchmarks, particularly in more complex microservice-satellite patterns. This superior performance can be attributed to the algorithm's ability to leverage the increased number of service routing paths and nodes, enabling more advanced route strategies. The expanded deployment range provides greater flexibility in service selections and routing decisions, allowing the algorithm to identify more energy-efficient configurations. By contrast, the energy costs associated with routing paths generated by random and comp-greedy methods occasionally increased as the number of replicas grew. This counterintuitive behavior can be attributed to inefficient routing decisions caused by the higher number of service nodes and calling paths, which complicate the decision-making process and lead to suboptimal resource utilization.

6) *Adaptability and stability*: Furthermore, as depicted in Fig. 10, which presents the energy distribution derived from 300 independent experiments, underscores the superior stability of the Steiner policy compared to other benchmark strategies. The results reveal that, even under sophisticated treed topologies with heightened complexity, the Steiner policy exhibits less variance in energy costs. This stability highlights its robustness and adaptability in managing diverse network conditions and service requirements.

VIII. CONCLUSION

In this paper, we proposed a DST-based solution to provide an energy-efficient service routing scheme for tasks from terrestrial clients. We systematically developed the SCPN models, including communication and computation, covering various aspects of the LEO satellite network. We also mathematically formulated the energy optimization problem for satellite-microservice routing and transformed it into finding a feasible DST solution within an augmented directed satellite-microservice colocated graph. To approximate the optimal solution, we designed a series of heuristic algorithms. Experimental results show that our algorithm significantly outperforms other benchmarks in energy efficiency, robustness, and adaptability across different network and microservice configurations in the context of SCPN.

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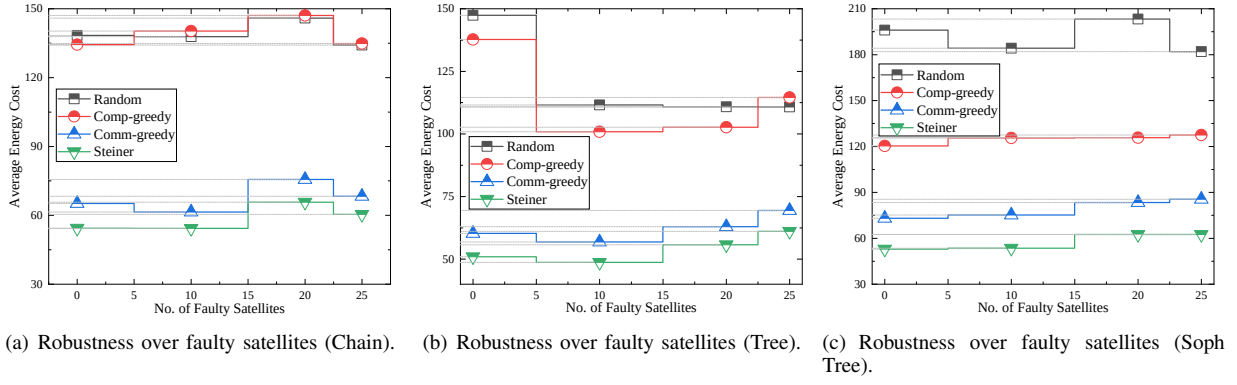


Fig. 8: Performance comparison on energy-cost under different configurations on number of faulty satellites for dynamic service request topologies.

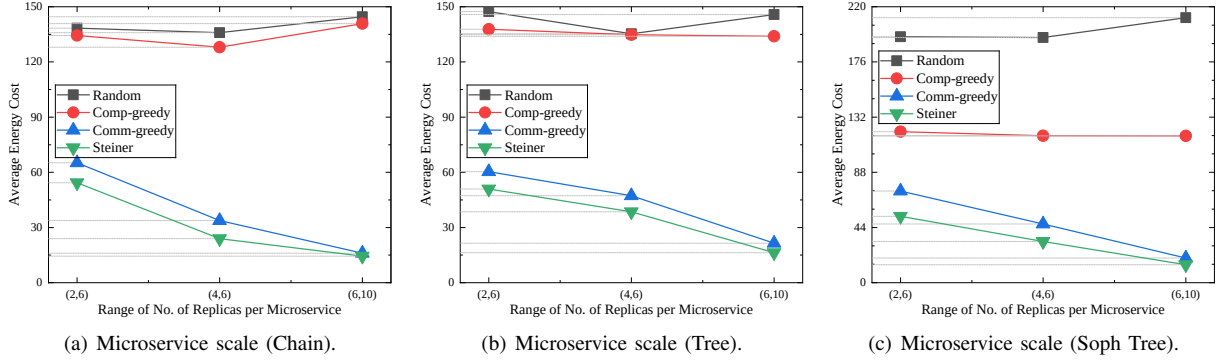


Fig. 9: Performance comparison on energy-cost under different configurations on number of replicas of microservices for dynamic service request topologies.

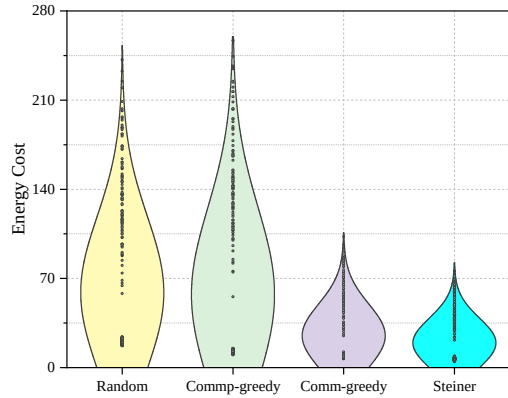


Fig. 10: Energy distribution of Soph Tree.

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